

Homework #1

Q 3.12

- A Weak Entity is an entity that is dependent on another. Example given in class was Room 117 is dependent on Hayes Hall (which is in turn dependent on BGSU). An owner entity type that has the key attribute for a weak attribute (likely a Regular entity type). The identifying relationship is the relationship that the Weak Entity has with it's Owner Entity, in the Example given it'd be that Hayes Hall has rooms (containing 117). A Partial key is an attribute that identifies unique weak entity that are related to the same owner type, so in the example, it'd be the room number "117" instead of 118 and so on.
 - Regular – has a key attribute
 - Weak entity type – Does not have a key attributes
 - Owner / identifying identity type – Has the key to identify the weak entity
 - Identifying relationship – The relationship that relates a weak entity type to its owner.
 - Partial Key – Attribute that can uniquely identify weak entities that are related to the same owner entity

Q 3.27

Entity 1	Cardinality Ratio	Entity 2
Student	1 to 1 (Social Security # is unique per person would only be higher if it means just the card or numbers from multiple countries)	Social Security Card
Student	Many to many Students have multiple classes, and teachers also have many students	Teacher
Classroom	Many to Many (Walls can be shared between rooms, ex. Room 117 and 118)	Wall
Country	1 to 1 (I am unaware of any country that has multiple Presidents, or someone	Current President

	who has been the current president of multiple countries outside of Empire)	
Course	Many to Many (Courses can have multiple books and books can be used in many courses)	Textbook
Item	Many to Many (1 item can be in many orders, and those orders can have many items)	Order
Student	Many to Many (Same as teacher, but students can have multiple classes, classes have many students)	Class
Class	Many to 1 (Assuming it means section, then a class only has one Instructor, possibly TAs)	Instructor
Instructor	1 to 1 (Unless offices are shared, an instructor should have their own office and the office would only have 1 instructor)	Office
eBay Auction Item	1 to Many (Items on eBay are unique, so there can be many bids on a unique item, but there can't be a singular bid on many Auctions).	eBay Bid